Reading Data

IMF quota data

I'm going to start by reading in the raw quota data before reading in the other data. I plan to clean them separately then attempt to combine them.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
# Reading in IMF quota data

raw_quota_data <-
  read.csv("raw_quota_data.csv")

head(raw_quota_data)
```

```
  Table.1..Updated.Quota.Formula.Variables.Shares.1.            X X.1
1                                        (In percent)              NA
2                                                     14th Review   NA
3                                                                   NA
4                                      United States       17.392  NA
```

```
5                                          Japan        6.458  NA
6                                       China 2/        6.388  NA
                   X.2      X.3 X.4     X.5      X.6 X.7     X.8      X.9 X.10
1                                   NA                   NA                   NA
2 CQS (Current Formula)             NA MER GDP           NA PPP GDP           NA
3               Current Previous  NA Current Previous  NA Current Previous   NA
4                14.923   14.775  NA  25.601   25.178  NA  15.252   15.219   NA
5                 3.986    4.345  NA   4.451    5.096  NA   3.561    3.694   NA
6                14.455   14.611  NA  18.061   18.350  NA  19.046   19.066   NA
      X.11     X.12 X.13     X.14     X.15 X.16        X.17     X.18 X.19
1                     NA                     NA                         NA
2 GDP Blend           NA Openness            NA Variability             NA
3   Current Previous   NA  Current Previous   NA     Current Previous   NA
4    21.462   21.195   NA   13.167   13.167   NA      10.540   10.246   NA
5     4.095    4.535   NA    3.587    3.699   NA       3.029    3.780   NA
6    18.455   18.636   NA   11.562   11.641   NA      11.867   12.437   NA
      X.20     X.21 X.22
1                    NA
2 Reserves           NA
3  Current Previous   NA
4    2.042    2.004   NA
5   10.004   10.362   NA
6   26.953   26.375   NA
```

```
tail(raw_quota_data)
```

```
194
195
196
197 1/ Data columns marked as "current" refer to the 2025 quota data update (data through 202
198                                                                          2/ Including
199
          X X.1     X.2     X.3 X.4     X.5     X.6 X.7     X.8     X.9 X.10
194 0.00052  NA 0.00035 0.00034  NA 0.00006 0.00006  NA 0.00003 0.00004   NA
195          NA                  NA                  NA                  NA
196          NA                  NA                  NA                  NA
197          NA                  NA                  NA                  NA
198          NA                  NA                  NA                  NA
199          NA                  NA                  NA                  NA
      X.11    X.12 X.13    X.14    X.15 X.16    X.17    X.18 X.19    X.20
194 0.00005 0.00005   NA 0.00023 0.00024   NA 0.00060 0.00053   NA 0.00064
```

```
195                    NA                    NA                    NA
196                    NA                    NA                    NA
197                    NA                    NA                    NA
198                    NA                    NA                    NA
199                    NA                    NA                    NA
       X.21 X.22
194 0.00066   NA
195           NA
196           NA
197           NA
198           NA
199           NA
```

The quota data looks really messy because it was from an Excel spreadsheet with empty rows and values that I think were put in stylistically. There's also cases of what should be column names being values. When I clean this data for my own uses, I'll only need 2 columns: country name, and CQS, which refers to the country's quota. I'll also need to make sure that column names are separated from values. I'll also want to make sure the names of countries are standardized with the names of countries in the other dataset I am using. Then on top of that I'll need to run tests to ensure that the values themselves actually make sense based on what I'm studying.

IMF conditionality data

Now I'm going to read in the raw IMF conditionality data.

```
# Reading in IMF conditionality data

raw_conditionality_data <-
  read.csv("raw_IMF_conditionality_data.csv")

head(raw_conditionality_data)
```

```
  Country.Name Country.Code Arrangement.Date Arrangement.ID Arrangement.Type
1  Afghanistan          AFG      26-Jun-2006      EBM/06/58              PRGF
2  Afghanistan          AFG      26-Jun-2006      EBM/06/58              PRGF
3  Afghanistan          AFG      26-Jun-2006      EBM/06/58              PRGF
4  Afghanistan          AFG      26-Jun-2006      EBM/06/58              PRGF
5  Afghanistan          AFG      26-Jun-2006      EBM/06/58              PRGF
6  Afghanistan          AFG      26-Jun-2006      EBM/06/58              PRGF
  Arrangement.Duration Arrangement.Amount Condition.Type
```

```
1                   36                 81              IB
2                   36                 81             QPC
3                   36                 81             QPC
4                   36                 81             QPC
5                   36                 81             QPC
6                   36                 81             QPC
              Condition.Text Condition.Year Condition.Month
1             Fiscal deficit           2006               .
2      Medium/long-term debt           2006               .
3            Short-term debt           2006               .
4             No new arrears           2006               .
5    Currency in circulation           2006               .
6 Net international reserves           2006               .
  Condition.Number.Applicable Condition.Policy.Area Condition.Source.Document
1                           3                   FP                 EBS/06/66
2                           3                  DEB                 EBS/06/66
3                           3                  DEB                 EBS/06/66
4                           3                  DEB                 EBS/06/66
5                           3                  EXT                 EBS/06/66
6                           3                  EXT                 EBS/06/66
  Condition.Implementation.Status Condition.Waiver
1                               .               NA
2                               .               NA
3                               .               NA
4                               .               NA
5                               .               NA
6                               .               NA
  Condition.Waiver.Source.Document Additional.Notes
1
2
3
4
5
6
```

```
tail(raw_conditionality_data)
```

```
      Country.Name Country.Code Arrangement.Date Arrangement.ID
35910     Zimbabwe          ZWE      02-Aug-1999      EBM/99/86
35911     Zimbabwe          ZWE      02-Aug-1999      EBM/99/86
35912     Zimbabwe          ZWE      02-Aug-1999      EBM/99/86
35913     Zimbabwe          ZWE      02-Aug-1999      EBM/99/86
```

```
35914     Zimbabwe          ZWE      02-Aug-1999      EBM/99/86
35915     Zimbabwe          ZWE      02-Aug-1999      EBM/99/86
      Arrangement.Type Arrangement.Duration Arrangement.Amount Condition.Type
35910              SBA                   14             141.36            SPC
35911              SBA                   14             141.36            SPC
35912              SBA                   14             141.36            SPC
35913              SBA                   14             141.36            SPC
35914              SBA                   14             141.36            SPC
35915              SBA                   14             141.36            SPC

35910
35911
35912
35913 The sale of 30 percent of ZIMRE shares held by the government; the sale of all subsidi
35914
35915
meal to be sold at a controlled retail price no lower than Z$393 per 50 kilo bag
      Condition.Year Condition.Month Condition.Number.Applicable
35910           1999               .                          1
35911           1999               .                          1
35912           1999               .                          1
35913           1999               .                          1
35914           1999               .                          1
35915           1999               .                          1
      Condition.Policy.Area Condition.Source.Document
35910                   FIN               EBS/99/131
35911                   LAB               EBS/99/131
35912                   LAB               EBS/99/131
35913                   PRI               EBS/99/131
35914                   PRI               EBS/99/131
35915                    SP               EBS/99/131
      Condition.Implementation.Status Condition.Waiver
35910               No Info (Pre-09)               NA
35911               No Info (Pre-09)               NA
35912               No Info (Pre-09)               NA
35913               No Info (Pre-09)               NA
35914               No Info (Pre-09)               NA
35915               No Info (Pre-09)               NA
      Condition.Waiver.Source.Document Additional.Notes
35910
35911
35912
35913
```

```
35914
35915
```

This data looks a lot better than the other dataset. It was also originally an Excel file, but the data is properly arranged in the rows and columns. To clean this dataset, I'll need to get rid of some of the columns I'm not interested in, leaving me only with: country name, condition type, and condition policy area. (I'm considering keeping condition text to maybe analyse word count, but I'm not sure yet). One of the issues I have is that this dataset lists each loan condition as a separate row, when what I actually want is one row per country that lists the number of of total conditions, and the number of each condition type and condition policy area. This will be more complicated than I had originally imagined, where I had originally planned to just convert the data frame from tall to wide format. I think I'll need to use some sort of sorting function to count number of variables and make a new data frame that I can actually use for my analysis, and for combining with the other dataset. Another issue is that this dataset, since it's organized by loans, only includes countries that have IMF loans. I want to analyze the relationship between voting power and loans, so I'll need to figure out a way to account for missing countries when I combine this with my other dataset. Once I sort this out, I'll have to do all the regular testing to make sure values make sense.

Colonization data

I haven't yet decided to which colonization dataset I'll use. I was originally going to use COLDAT, but it only shows if a place was colonized by Europe and not North America, which I think is problematic, particularly because of the US and Japan. I am going to see if I can find a simple dataset that has what I need, and if not, I may either have to abandon this or somehow add my own column where I manually fill in values.

Combining different data sets

I'm hoping that once I clean my data separately, it shouldn't be too much work to combine them. I plan to attach the conditionality dataset onto the quota dataset using some sort of mutate function. The biggest issue I can foresee is that there are countries in the quota dataset who are not present in the conditionalities dataset because they don't have loans. This might be challenging as they will have missing values. I should be able to fix this by changing all missing values in those columns to 0 (for no loan).